

KIRUBEL

SECURITY RESEARCHER · SYSTEMS ENGINEER · SOFTWARE DEVELOPER

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PROFILE

Computer science student at the American College of Technology with a strong focus on offensive security, low-level systems, and software engineering. Independent builder of open-source security tooling – from runtime SBOM generators and ELF instrumentation engines to kernel-space eBPF monitors and hardware-accelerated threat scanners. Comfortable across C, C++, Assembly, and modern application stacks. Active in vulnerability research with published work in binary fuzzing and media-engine analysis.

EDUCATION

American College of Technology (ACT)

Present

Addis Ababa, Ethiopia – Undergraduate Program

Coursework in computer science foundations, programming, data structures, operating systems, and networking, alongside independent research in systems security.

PROJECTS

SAUR – Security Analysis & Threat Engine

C++ / YARA

Hardware-accelerated system auditor that scans files and memory for bad actors, credential leaks, and malicious payloads using AVX2 vector instructions, YARA rulesets, and live URLhaus indicator feeds.

> C++, AVX2, RE2, YARA, Linux

K-Sentinel – Kernel-Space Execution Monitor

C++ / eBPF

Kernel-space security engine using eBPF to monitor and block malicious process execution (execve) with sub-microsecond latency. Operates entirely from the kernel without user-space agents.

> C++, eBPF, Linux Kernel

ZeroShadow – Control-Flow Integrity Supervisor

C++ / ptrace

Low-level Linux user-space supervisor enforcing coarse-grained Control-Flow Integrity. Single-steps target threads via ptrace, validating the instruction pointer against known-executable memory boundaries to detect anomalous execution.

> C++, ptrace, ELF, Linux

AegisPQC – Hybrid Post-Quantum Cryptography Framework

C++

Security framework combining classical encryption with quantum-safe encryption so a secure connection remains intact even if one algorithm is broken. Provides an abstraction layer allowing developers to swap cryptographic primitives without rearchitecting the application.

> C++, Post-Quantum Crypto, Hybrid KEM

dSBOM – Runtime SBOM Generator for Linux

C++

Non-invasive runtime Software Bill of Materials engine. Derives component identity, linkage, and cryptographic integrity assertions natively from the Linux kernel VFS and tracer subsystems, eliminating userspace injection, application instrumentation, or static manifest approximation.

> C++, Linux VFS, Tracer Subsystems

KASCVE – Kernel Asset Surface & CVE Evaluator

Python

Automated security auditing framework that evaluates kernel attack surface against known CVEs and the OWASP Top 10, supplemented by active penetration tests such as SQL injection detection. Shipped for both desktop and Android.

> Python, CVE Database, OWASP Top 10

Threat-Matrix – Edge-Native Threat Intelligence

TypeScript

Serverless dashboard aggregating live CVEs, exploits, and security news into a single stream. Built on Cloudflare Workers and D1 for sub-millisecond latency, with a zero-trust anonymous authentication model and no ad tracking. Companion Android client built in Kotlin.

> TypeScript, Cloudflare Workers, D1, Kotlin

rlottie Zero-Day Research – Fuzzing Harness

CMake

Independent vulnerability research on the rlottie animation engine. Engineered an AFL++ fuzzing harness executing systematic crash analysis, isolating memory corruption, input-validation failures, and memory leaks affecting third-party messaging clients.

> C++, AFL++, ASAN, CMake

KiOS – Kernel-Integrated Secure Linux

Shell / Python

Hardened Arch-based Linux distribution engineered from the kernel up to reduce attack surface while maintaining developer velocity. Includes custom automation for localization, provisioning pipelines, and cryptographic storage layouts.

> Bash, Python, Arch Linux

CHAOS – Procedural 3D Web Engine

HTML / Three.js

Self-contained Three.js application manifesting a unique 3D visual dimension on every load, driven by a seed-based procedural generator and rendered entirely in a single HTML file.

> Three.js, WebGL, JavaScript

hlinux – Linux-in-Browser Runtime

HTML / v86

A full Linux environment running inside the browser using v86 emulation, supporting shell interaction without any server-side component.

> v86, WebAssembly, JavaScript

SKILLS

Languages: C, C++, C#, Assembly (x86 / ARM), Python, Go, Rust, Kotlin, TypeScript, JavaScript, Bash, CMake

Security: AFL++, Ghidra, IDA Pro, GDB / ptrace, Burp Suite, Nmap, Metasploit, Wireshark, YARA, eBPF

Frameworks: React, Next.js, Node.js, FastAPI, Three.js, PostgreSQL, SQLite

Systems: Arch Linux, Debian, Docker, Git / CI pipelines, CMake, Vim

AREAS OF INTEREST

- Offensive security – exploit development, fuzzing, binary analysis.
- Low-level systems – kernel internals, tracing, memory safety.
- Post-quantum cryptography and secure protocol design.
- Tooling – SBOM, supply-chain security, and defensive automation.

- Digital design – 3D graphics, procedural art, and interactive web.